

# Mohamed Ahmed Rabea

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## PROFILE

Mid-Level Unity Developer with a Computer Engineering degree and a strong background in programming and AI. Skilled in C#, Unity, Object-Oriented Programming, and VR tools, with hands-on experience in game development and immersive projects. Actively contributing to real-world projects by designing gameplay features, collaborating with cross-functional teams, and setting up tasks to support smooth development workflows. Motivated to grow as a developer while bringing creative ideas into engaging player experiences.

## EDUCATION

### **Benha University | Cairo, Egypt**

Shoubra faculty of engineering

**Key Modules:** programming, Maths & statistics, Databases, AI, python, C++, OOP, C#, Graphics.

**Grade:** outstanding marks during my five years with an A Grade.

**September 2020 – June 2025**

## EXPERIENCE

### **Genesis Creations**

#### **Mid-Level Unity Developer**

- Developed VR experiences and interactive games using Unity.
- Built gameplay systems and implemented modern tools and packages to enhance functionality.
- Optimized performance and memory usage to improve player experience across platforms.
- Collaborated with teammates from different disciplines to ensure smooth development of workflows.
- Organized and assigned tasks using Click Up to support project management and team productivity.

**February 2026 – Present**

### **Genesis Creations**

#### **Unity Developer intern**

- Supported development of VR experiences and game features.
- Assisted in implementing gameplay mechanics and UI elements.
- Learned and applied Unity best practices, debugging, and optimization basics.
- Contributed to team tasks and gained hands-on experience in production workflows.

**July 2025 – January 2026**

### **Technovate, IEEE, and A-One Training**

#### **AI Instructor**

- Taught foundational and advanced AI concepts to engineering and computer science students.
- Designed and delivered **hands-on workshops** using Python, TensorFlow, and real-world datasets.
- Created structured learning paths covering algorithms, neural networks, and model evaluation.
- Guided students through building **projects**, including image classification, NLP, and data pipelines.
- Simplified complex AI concepts into **beginner-friendly sessions** for mixed-level audiences.
- Mentored students on academic and career development in AI and software engineering.
- Coordinated and taught sessions for **IEEE AI Bootcamp** with 60+ participants.

**September 2024 – February 2025**

### **National Telecommunication institute (NTI) | Cairo, Egypt**

#### **AI Trainee**

- I learned about Machine learning and Deep learning and how to implement their models after wrangling data.
- I learned about soft skills including communication, Leadership, Time management, Presentation making.

**July 2023 – September 2023**

## PROJECTS

### Ship Simulation

#### *A VR Project*

- Developing a realistic **ship maneuvering simulation** using Unity and C#.
- Implementing core ship systems including **propulsion, steering, and environmental interactions**.
- Developed navigation scenarios such as docking, undocking, and harbor maneuvering.
- Integrated advanced systems including **multibeam sonar** for underwater mapping and depth visualization.
- Optimizing performance for real-time simulation with large environments and dynamic systems.

### Afrexim Bank Crisis Simulation

#### *A VR Project*

- Developed a **digital twin VR experience** for Afrexim Bank using Unity and C#.
- Designed interactive scenarios to demonstrate **the workflow done when crisis happens**.
- Implementing optimized real-time interactions and smooth navigation within VR.
- Ensuring performance, stability, and usability across VR platforms.

### Sterile Compounding simulation

#### *A VR Project*

- Developed an interactive VR environment using **Unity, C#, and Oculus frameworks**.
- Built modular gameplay systems (Cleaning system, interaction handlers, objective tracking).
- Implemented **player movement, hand tracking, and object interaction** using Oculus Integration.
- optimized **physics-based interactions, and UI systems** for VR performance constraints.
- Integrated **event-driven systems and scriptable objects** to enhance scalability and maintainability.
- Optimized scene performance through **baked lighting, asset management**, and profiling tools.

### Dubai Airport simulation

#### *A VR and PC Project*

- Developed an interactive **digital twin of Dubai International Airport** using Unity and C#, enabling real-time visualization of airport operations.
- Implemented **accurate object behaviours, and system logic** reflecting real airport processes (gates, vehicles, Racks).
- Built modular systems for **navigation, state management, and dynamic data visualization**.
- Optimized large-scale airport scenes through **LOD systems, light baking, occlusion culling**.
- Collaborated on **integration with operational data** to simulate realistic airport scenarios.
- Ensured high performance and stability for complex environments with thousands of active objects.

### Explainable AI for Medical Diagnosis from Radiographs

#### *Graduation XAI Project*

- **Developed an end-to-end Explainable AI (XAI) diagnostic system** for classifying diseases from (CXR).
- Engineered a high-performance DL solution based on the **(ViT-Base-R50-S16-224)** architecture.
- Achieved robust test performance on the NIH ChestX-ray14 dataset.
- Implemented a lightweight fine-tuning strategy to optimize generalization and reduce overfitting.
- Pioneered a **pathology-specific XAI approach**: utilized **SHAP, Attention Rollout, GradCAM++**.
- Created a **thorax-Extraction preprocessing pipeline using OpenCV** to isolate lung regions.
- Deployed the full system as an interactive, user-friendly **web-based platform (FastAPI)**.

### Kitchen Chaos

#### *A Game Project*

- Made a 3d cooking Game (Overcooked style) using unity with Well-Designed Generic OOP Code for learning.

### Snake Game

#### *A Game Project*

- Designed and a classic Snake Game using P3, focusing on smooth gameplay mechanics and responsive controls.

## AI Chess Game

### A Game Project

- Made a Chess Game Using Java GUI libraries and Well-Designed Generic OOP Code.
- Implemented an AI algorithm so that the player can play against the AI.

## Responsive restaurant website

### A website Project

- Made a responsive website for a fancy Restaurant using HTML, SCSS, JavaScript.

## Technical SKILLS

- **Languages:** Arabic (*Native*), English (*Fluent*), German (Beginner)
- **Engine:** Unity (2D, 3D).
- **OOP:** familiarity with Design patterns, writing good clean code in C#.
- **Programming:** C#, Python, Java and Problem-solving skills using C++, OpenGL.
- **AI:** Machine learning, Deep learning, Data Preparation.
- **Mathematics & Physics:** Linear Algebra, Trigonometry, Kinematics.
- **2D Art:** Photoshop and Illustrator.
- **3D Modelling:** familiarity with Blender.
- **Audio:** Familiarity with Audacity.
- **Web Development:** HTML, CSS, SASS, JavaScript, React, SQL and Database Concepts.
- **Tools:** Git, Fork, Figma, Click Up, Postman.
- **Platforms:** PC, VR.

## Activities

- Attended gaming events (Egypt Game Jam, ITI Game Day, Arabic Games Conference, etc)